Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions for a system to be in static equilibrium.

Vector sum of forces acting on the system = 0 [1]

Sum of clockwise moments about a pivot = sum of anticlockwise moments about the same pivot. [1]

.....[2]

(b) A uniform wooden rod of length 0.30 m and mass 0.45 kg is attached to a vertical wall with a hinge at an angle of 40° with the horizontal. The other end of the rod is attached to the wall with a rope, so that the rope makes an angle of 30° with the horizontal. An object M of mass 1.2 kg is attached to the other end of the rod, as shown in Fig. 1.1 below.

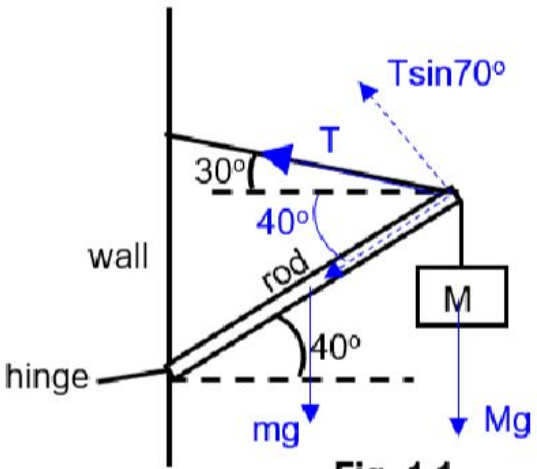


Fig. 1.1

(i) Show that the tension in the rope is 11 N.

Let length of rod be L.
Take moments about hinge.
Anticlockwise moment = clockwise moment

$$T\sin 70^\circ \times L = \left(Mg \times L\cos 40^\circ\right) + \left(mg \times \frac{L}{2}\cos 40^\circ\right)$$

$$T = \frac{\left(M + \frac{m}{2}\right) g\cos 40^\circ}{\sin 70^\circ}$$

$$= \frac{\left(1.2 + \frac{0.45}{2}\right) \times 9.81 \times \cos 40^\circ}{\sin 70^\circ}$$

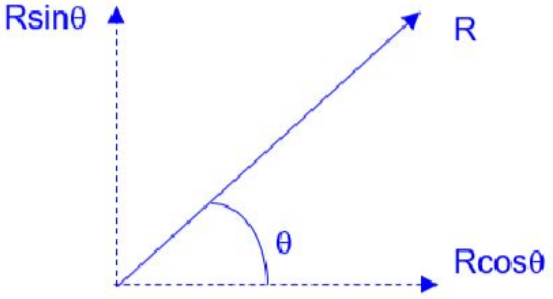
$$= 11.395$$

$$\approx 11\ \text{N}$$

[1 for correct anticlockwise moment; 1 for correct total clockwise moment
1 for more sf before final answer]

[3]

(ii) Determine the magnitude of the reaction force from the hinge acting on the rod, and the angle that the reaction force makes with the horizontal.



Take upwards and rightwards to be positive.

Let reaction force be R

Horizontally

$$R\cos\theta = T\cos 30^\circ$$
$$= 11\cos 30^\circ$$
$$= 9.526 \text{ N} \quad (1) \quad [1 \text{ mark}]$$

Vertically

$$R\sin\theta + T\sin 30^\circ = (M+m)g$$
$$R\sin\theta = (M+m)g - T\sin 30^\circ$$
$$= \left[(1.2+0.45)\times 9.81\right] - 11\sin 30^\circ$$
$$= 10.69 \text{ N} \quad (2) \quad [1 \text{ mark}]$$

2(1) $\tan \theta = \dfrac{10.69}{9.526}$

$\theta = 48.30^\circ$

$\approx 48^\circ$ [1 mark]

Sub into (1) $R\cos 48.30 = 9.526$

$R = 14$ N [1 mark]

magnitude = ... N

direction = ... ° from the horizontal

[4]

[Total: 9]